LECTURE NOTES

Lecture 8:

AI504 - Knowledge Represntation

Simon Holm
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SDU
DEPARTMENT OF MATHEMATICS
AND COMPUTER SCIENCE

Contents

1 Relative clauses

The logic of $\mathcal{A}(RC)$ of all $\_ + \underline{\text{relative}}$ clauses)

Previously, a **signature** was simply a set of nouns, now our signature consists of a set of nouns, **plus** a set of tranitive verbs (a verb must take an object).

- **What are sentences? What are models? What are proofs?**

1.1 Models

A Model (*relative to a particular signature*) consists of a set (*the domain*), and interpretations of both all nouns and all verbs

- $[\![]\!]_{\text{nouns}}$ is a subset of the domain
- $[\![]\!]_{\text{verbs}}$ is a binary relation on the domain

so..

$$[\![]\!]_{\text{nouns}} : P \to \mathcal{P}(M), \quad [\![]\!]_{\text{verbs}} : R \to \mathcal{P}(M \times M)$$

Where P is the set of nouns, and R is the set of verbs

so..

$$\mathcal{M} = (P, [\![]\!]_{\text{nouns}} : P \to \mathcal{P}(M), \quad [\![]\!]_{\text{verbs}} : R \to \mathcal{P}(M \times M))$$

1.2 Terms

We want to give meaning to relative clauses so we ignore irrelevant words.

$$\cancel{\text{those who}}\ \text{love all}\ \cancel{\text{who}}\ \text{fear all woodchucks}$$

$$\text{love all (fear all woodchucks)}$$

A term is either
- A single noun, or
- an expression (r all t), where r is a verb and t is a term

Terms

i

if t is a single noun, then
$[\![t]\!]$ is simply $[\![t]\!]_{\text{noun}}$, which comes from the data in $\mathcal{M}$

ii

if t is r all t', then
$$[\![t]\!] = \{m \in M : \forall n \in [\![t']\!] \quad (m, n) \in [\![r]\!]\}$$

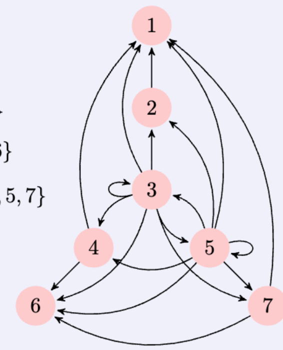
1.2.1 Example

Consider the model $\mathcal{M}$ with universe $\{1, \ldots, 7\}$, and with interpretations of three nouns, hawks, birds, and turtles, and one verb, see, shown below.

$[\![\text{hawks}]\!] = \{1\}$

$[\![\text{birds}]\!] = \{1, 6\}$

$[\![\text{turtles}]\!] = \{3, 5, 7\}$



In this model,

$$
\begin{array}{lcllcl}
[\![\text{see all hawks}]\!] & = & \{2,3,4,5,7\} & [\![\text{see all turtles}]\!] & = & \{3,5\} \\
[\![\text{see all birds}]\!] & = & \{3,4,5,7\} & [\![\text{see all (see all hawks)}]\!] & = & \{3,5\}
\end{array}
$$

Here are some examples of sentences false and true in the model:

$$
\begin{array}{l}
\mathcal{M} \not\models \text{All (see all hawks) (see all birds)} \\
\mathcal{M} \models \text{All (see all turtles) (see all (see all hawks))} \\
\mathcal{M} \models \text{All (see all (see all hawks)) (see all turtles)}
\end{array}
$$

Figure 1: Example 2.4 from: Logic From Language by Lawrence S. Moss

$$[\![\text{see all hawks}]\!] = \{m \in M : \forall n \in \{1,2,3,4,5,6\} \quad (m,n) \in \{1\}\} = \{2,3,4,5,7\}$$

$$[\![\text{see all birds}]\!] = \{m \in M : \forall n \in \{1,2,3,4,5,6\} \quad (m,n) \in \{1,6\}\} = \{3,4,5,7\}$$

$$[\![\text{see all turtles}]\!] = \{m \in M : \forall n \in \{1,2,3,4,5,6\} \quad (m,n) \in \{3,5,7\}\} = \{3,5\}$$

$$[\![\text{see all (see all hawks)}]\!] = \{m \in M : \forall n \in \{2,3,4,5,7\} \quad (m,n) \in \{1\}\} = \{3,5\}$$

1.3 Sentences

A **sentence** is an expression

$$\text{all term term}$$

1.4 Fundemental Lemma

Work within a fixed model.

For any terms s and t and any ver r. if $[\![s]\!] \subseteq [\![t]\!]$

$$[\![r \text{ all } t]\!] \subseteq [\![r \text{ all } s]\!]$$

1.4.1 Proof

Suppose that m is an arbitrary element of M.

Assume that $m \in [\![r \text{ all } t]\!]$ Goal: $m \in [\![r \text{ all } s]\!]$

This means that $\forall n \in [\![t]\!] \quad (m, n) \in [\![r]\!]$ Goal: $\forall n \in [\![s]\!] \quad (m, n) \in [\![s]\!]$

Now let n be an arbitrary element of $[\![s]\!]$ Goal: $(m, n) \in [\![s]\!]$

Since $[\![s]\!] \subseteq [\![t]\!]$, $n \in [\![t]\!]$

Now because of $\forall n \in [\![s]\!] \quad (m, n) \in [\![s]\!]$ and $n \in [\![t]\!]$

We know that $(m, n) \in [\![r]\!]$ $\square$